SINGULAR GAUSSIAN CONDITIONING IN HILBERT SPACE: INVERSE-FREE SCHUR PROJECTION AND VARIANCE COLLAPSE

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ABSTRACT. We study singular coordinate conditioning for Gaussian measures on separable Hilbert spaces. Since trace-class covariance blocks are compact, the usual Schur-complement formula cannot be implemented through a bounded inverse. We introduce an inverse-free variational Schur projection on the Cameron–Martin range, derive the canonical λ^{-2} target-variance scale by canonical scaling, and prove an $O(\lambda^{-6})$ residual estimate. The construction is stable under Galerkin truncation for exponentially stable analytic semigroup dynamics and applies to sectorial elliptic Gaussian fields.

1. Introduction

Singular coordinate conditioning arises when an exact constraint $X_k = x_k^*$ is reached as a zero-noise or infinite-precision limit of Gaussian conditioning problems. Finite-dimensional Gaussian conditioning is governed by the ordinary Schur complement of a covariance matrix [14, 20]. Infinite-dimensional Gaussian conditioning is subtler: Radon Gaussian measures on separable Hilbert or Banach spaces have compact covariance operators, and their Cameron–Martin spaces are covariance-energy domains rather than ambient statistical parameter spaces [2, 8, 22]. In Hilbert-space Gaussian conditioning, shorted operators give the covariance of conditioning on closed subspaces, while recent inverse-problem work emphasizes that positive-noise posterior formulas must be interpreted through approximation, disintegration, or Hilbert-space posterior equivalence rather than through a global covariance inverse [19, 23, 6, 4, 5].

The obstruction studied here is the covariance-block obstruction behind that phenomenon. In a finite-dimensional model, the conditional variance of a coordinate can be written as $A - BD^{-1}B^*$ after a block decomposition. In a separable Hilbert space the corresponding block D is typically compact and trace class, hence D^{-1} is not a bounded operator on the ambient space [2, 21, 8]. Classical Schur-complement and shorted-operator theory, originating in Krein's theory of non-negative extensions and developed systematically by Anderson and Trapp, provides variational tools for positive

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block operators [16, 1, 9]. However, this theory does not by itself give the scaling law for a singular Gaussian conditioning residual [24, 7, 11, 13]. This paper formulates the residual directly as a Cameron–Martin energy projection and never invokes a pseudoinverse or an ambient precision operator.

The hard-conditioning limit also reaches the Feldman–Hájek measure-class boundary. With positive observational noise, Gaussian Bayesian inverse problems may remain in the equivalent-measure regime [23, 4, 5]. Under an exact coordinate constraint, the limiting conditional law is supported on a hyperplane of prior measure zero, so the equivalent-measure regime is left [10, 12, 2]. A useful theory must therefore separate the diverging joint divergence caused by exact evidence from the finite covariance residual that remains visible in the block structure.

We prove that this finite residual is controlled by an inverse-free variational Schur projection. Under the Lyapunov block assumptions stated below, the target residual has the exact form

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad R_\lambda = O(\lambda^{-6}).$$

Consequently,

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}).$$

The proof uses covariance-energy forms, shorting, Douglas range inclusion, Lyapunov block identities, and trace-ideal convergence. These are standard inputs in operator and SPDE analysis [16, 1, 9] [21, 8]. The Cameron–Martin space is used only as a Hilbert subspace equipped with its covariance norm; no differentiable statistical or geometric structure is assumed [2]. The operator base is an exponentially stable analytic semigroup with bounded drift perturbations, so dissipative sectorial generators such as $-(-\Delta + \kappa^2)$ fall inside the abstract setup.

The paper has four main contributions:

(1) **Canonical normalization.** We identify the block-Lyapunov mechanism that fixes the observed coordinate variance at the unique finite-part scale $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$.
(2) **Inverse-free Schur projection.** We identify the residual subtraction with the target-coordinate quadratic form of the shorted-operator variational formula, then prove the $O(\lambda^{-6})$ residual estimate in the same operator-theoretic regime where compact covariance blocks preclude a bounded Schur inverse.
(3) **Analytic-semigroup and Galerkin stability.** We work with an analytic semigroup drift base plus bounded perturbations, and prove that finite-dimensional truncations converge to the Hilbert-space residual in trace ideal norm, matching the discretization-independent perspective used in Hilbert-space Gaussian inverse problems [23, 4, 5].

(4) **Radon Gaussian conditioning.** For Radon Gaussian laws, we construct the conditional predictor as a measurable Cameron–Martin projection and show that the associated renormalized conditioning functional remains bounded, using regular conditional probabilities and continuous Gaussian disintegration [2, 17, 22].

1.1. **Positioning in the literature.**

The paper is closest to four bodies of work. First, it uses the classical measure-class structure of Gaussian measures, especially the Cameron–Martin theorem and the Feldman–Hájek dichotomy, as presented in standard references on Gaussian measures [10, 12, 2]. Second, it is adjacent to Bayesian inverse problems and Hilbert-space Gaussian conditioning, including the shorted-operator description of Gaussian conditioning on closed subspaces, positive-noise posterior formulas, low-rank posterior approximations, and nonlinear observation maps [19, 23, 6, 22, 4, 5]. Third, the variational residual is informed by Schur complements, shorted operators, Douglas range inclusion, and modern complementability theory for block operators [16, 1, 9, 24, 7, 11, 13]. Fourth, the motivating examples come from stochastic evolution equations and elliptic Gaussian fields, where dissipative sectorial generators produce compact trace-class stationary covariances that are naturally approximated by Galerkin truncation [8, 18, 21].

1.2. **Organization of the paper.**

Section 2 defines the operator model and establishes the canonical normalization. Section 3 establishes the Hilbert-space variational Schur residual. Section 4 gives the Radon Gaussian conditional extension and the renormalized conditioning functional. Section 5 gives an elliptic Gaussian-field example. Section 6 records limitations and extensions, and Section 7 concludes. Proofs are given in Sections A and B; numerical diagnostics and Galerkin verification are given in Section C.

2. Operator Model and Canonical Normalization

2.1. **Operator Dynamics and Regularized Conditioning.**

We model the continuous-time dynamics as stochastic evolution on a separable Hilbert space $\mathcal{H}$ [8]:

$$dX_t = \mathcal{A}X_t\,dt + \mathcal{D}^{1/2}dW_t \tag{1}$$

where $\mathcal{A}$ is the drift operator and $\mathcal{D}$ is the trace-class diffusion operator. We decompose the Hilbert space as $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$, where $\mathcal{H}_k := \mathbb{R}\phi_k$ is the one-dimensional target coordinate subspace, and $\mathcal{H}_I := \mathcal{H}_k^{\perp}$ is the free bulk subspace. Let $\Pi_k := \phi_k \otimes \phi_k$ and $\Pi_I := I - \Pi_k$ be the corresponding orthogonal projections.

To approximate the exact point conditioning $X_k = x_k^*$, we set the incoming target-coordinate blocks to zero and apply a regularization parameter $\lambda > 0$.

Definition 1 (Calibrated Approximating Dynamics Family)**.** The regularized conditioning operator of strength $\lambda > 0$ at coordinate X_k acts on the operator dynamics (1) by modifying both the drift and the diffusion:

$$\mathcal{A}^{(\lambda)} := \mathcal{A}_{\text{cond}} - \lambda\Pi_k, \qquad \mathcal{D}^{(\lambda)} := \mathcal{D}_{\text{cond}} + d_\lambda\Pi_k \tag{2}$$

where $\mathcal{A}_{\text{cond}}$ is the block drift operator with incoming target-coordinate blocks set to zero ($\Pi_k\mathcal{A}_{\text{cond}} = 0$), $\mathcal{D}_{\text{cond}}$ has target-coordinate cross-noise set to zero, and $d_\lambda > 0$ is the local diffusion variance of the target coordinate at regularization strength λ. In the SDE, this corresponds to:

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{d_\lambda}\, dW_{k,t}. \tag{3}$$

2.2. **Canonical Normalization.** The block-decoupled structure under the regularized dynamics yields:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} d_\lambda & 0 \\ 0 & D_{II} \end{pmatrix}, \tag{4}$$

where $a_{Ik} := \mathcal{A}_{Ik}\phi_k \in \mathcal{H}_I$ represents the off-diagonal drift coupling. Let Σ_λ denote the stationary covariance operator under regularization strength λ, solving the Lyapunov equation:

$$\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0. \tag{5}$$

The target block kk decouples completely from the free coordinates, yielding the exact scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + d_\lambda = 0, \qquad \Sigma_{\lambda,kk} = \frac{d_\lambda}{2\lambda}. \tag{6}$$

The normalization is fixed by canonical scaling, not by a free modeling convention. In the Gaussian finite-part extension below, a constant forcing b_k produces the deterministic target offset

$$\Delta_\lambda := m_{\lambda,k} - x_k^* = \frac{b_k}{\lambda} = O(\lambda^{-1}), \tag{7}$$

so the squared deterministic displacement has the rigid scale $\Delta_\lambda^2 = b_k^2/\lambda^2 = O(\lambda^{-2})$. The leading target contribution to the conditional finite-part functional is the signal-to-noise ratio

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2/\lambda^2}{2\Sigma_{\lambda,kk}}. \tag{8}$$

Consequently the target covariance must decay at the same order as the squared deterministic displacement. If $\Sigma_{\lambda,kk} = O(\lambda^{-1})$, as for the slow family $d_\lambda = \varepsilon_0$, then (8) is $O(\lambda^{-1})$ and the finite target contribution collapses to zero. If $\Sigma_{\lambda,kk} = O(\lambda^{-3})$, as for the fast family $d_\lambda = \varepsilon_0/\lambda^2$, then (8) is $O(\lambda)$ and the target contribution diverges. These two failures are the slow and fast ablations reported in Section C.2. A nonzero finite limit therefore forces the critical covariance scale

$$\Sigma_{\lambda,kk} \asymp \lambda^{-2}. \tag{9}$$

Proposition 2 (Canonical Scaling)**.** *The unique target-diffusion scaling that makes the deterministic offset and target variance balance at a finite nonzero scale is*

$$\Sigma_{\lambda,kk} \equiv \frac{\varepsilon_0}{2\lambda^2}, \qquad \varepsilon_0 > 0. \tag{10}$$

By the scalar Lyapunov identity (6)*, this is equivalent to*

$$d_\lambda = 2\lambda\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{\lambda} \quad \textit{for every } \lambda > 0. \tag{11}$$

With this canonical family,

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2}{\varepsilon_0}, \tag{12}$$

and the calibrated approximating dynamics become

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{\varepsilon_0/\lambda}\, dW_{k,t}. \tag{13}$$

All subsequent results are stated for this canonical scaling family.

Definition 3 (Schur LMMSE Invariant)**.** Let $S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I} X_k\|_{L^2}^2$ represent the linear minimum mean squared error (LMMSE) residual of the target coordinate X_k on the closed linear span $\mathcal{M}_I$ of the free coordinates X_I. The *Schur LMMSE invariant* m_2^{Schur} is defined as:

$$m_2^{\text{Schur}} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}. \tag{14}$$

2.3. **Operator Setup and Assumptions.** We state the operator-theoretic assumptions that make the limit passage independent of the Galerkin dimension.

Definition 4 (Drift Operator with Analytic Base)**.** A drift operator on a separable Hilbert space $\mathcal{H}$ is an operator of the form $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where:

- (i) $\mathcal{A}_0 : \text{dom}(\mathcal{A}_0) \subset \mathcal{H} \to \mathcal{H}$ is closed, densely defined, and dissipative, and it generates a strongly continuous analytic contraction semigroup $(e^{t\mathcal{A}_0})_{t\geq 0}$ on $\mathcal{H}$ satisfying $\|e^{t\mathcal{A}_0}\| \leq e^{-\alpha t}$ for some spectral gap $\alpha > 0$; equivalently, the base operator $-\mathcal{A}_0$ is sectorial of angle $< \pi/2$ in this exponentially stable contraction setting.
- (ii) $\mathcal{K} : \mathcal{H} \to \mathcal{H}$ is a bounded linear operator.
- (iii) The full operator $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, with $\text{dom}(\mathcal{A}) = \text{dom}(\mathcal{A}_0)$, generates a strongly continuous analytic semigroup $(e^{t\mathcal{A}})_{t\geq 0}$ by the bounded perturbation theorem, and this semigroup remains exponentially stable: $\|e^{t\mathcal{A}}\| \leq Me^{-\beta t}$ for some $M \geq 1$, $\beta > 0$.

In finite dimensions ($\mathcal{H} = \mathbb{R}^n$), this reduces to a decomposition of a Hurwitz drift matrix into an exponentially stable analytic base plus a bounded perturbation.

Definition 5 (Trace-Class Noise and Covariance)**.** Let $\mathcal{S}_1(\mathcal{H})$ denote the Banach space of trace-class (nuclear) operators on $\mathcal{H}$ equipped with the trace norm $\|T\|_{\mathcal{S}_1} := \text{Tr}(|T|) = \sum_{i=1}^\infty \langle \phi_i, (T^*T)^{1/2}\phi_i \rangle < \infty$.

(i) The diffusion operator $\mathcal{D} : \mathcal{H} \to \mathcal{H}$ is positive, self-adjoint, and belongs to $\mathcal{S}_1(\mathcal{H})$.

(ii) Assuming the pair $(\mathcal{A}, \mathcal{D}^{1/2})$ is controllable, the stationary covariance operator $\Sigma : \mathcal{H} \to \mathcal{H}$ is the unique strictly positive, self-adjoint, trace-class solution to the operator Lyapunov equation:

$$\mathcal{A}\Sigma + \Sigma\mathcal{A}^* + \mathcal{D} = 0 \tag{15}$$

which is given by the convergent Bochner integral

$$\Sigma = \int_0^\infty e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\,dt.$$

Assumption 6 (Core Analytical Assumptions for Singular Hyperplane Conditioning)**.** This work relies on two core analytical assumptions regarding the target-coordinate block decoupling and the outgoing coupling:

(i) **Linear Second-Moment Channel:** A centered linear stationary process on $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ has finite second moments and covariance solving Equation (15). The block-decoupling constraint zeroes all incoming target drift and cross-noise, so that $\mathcal{A}_{kI} = \mathcal{D}_{kI} = \mathcal{D}_{Ik} = 0$ and $\mathcal{A}_{kk} = -\lambda$. The target diffusion $\mathcal{D}_{kk} = \varepsilon_0/\lambda$ is the unique scaling determined by the canonical normalization (Proposition 2). The free noise is independent of the local target noise. (Gaussianity is not assumed).

(ii) **Cameron–Martin Stability of Off-Diagonal Drift Coupling:** Let $Q_0 := \Sigma_{II}^{(0)}$ denote the free-block covariance produced by the free noise under block decoupling, and let $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ with the energy norm $|u|_{Q_0} := \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. The outgoing vector $a_{Ik} := \mathcal{A}_{Ik}\phi_k$ belongs to H_{Q_0}, and $e^{t\mathcal{A}_{II}}$ restricts to a strongly continuous exponentially stable semigroup on H_{Q_0}:

$$\|e^{t\mathcal{A}_{II}}\|_{\mathcal{L}(H_{Q_0})} \le Me^{-\omega t}, \qquad t \ge 0, \tag{16}$$

for constants $M \ge 1$ and $\omega > 0$ independent of λ.

Intuitive Explanation: Part (i) states that the regularized dynamics act as a strong localized drift penalization on the target coordinate X_k while isolating it from incoming influences, giving an operator implementation of the block-decoupling constraint. Part (ii) requires that the off-diagonal drift coupling a_{Ik} from the target coordinate has finite energy relative to the background fluctuations of the free coordinates. This ensures that target fluctuations propagating downstream do not overload the variance of the infinite-dimensional system.

Remark 7 (Finite-Energy Outgoing Covariance Coupling and Finite-Dimensional Calibration). The membership $a_{Ik} \in H_{Q_0}$ is an admissibility condition, not a consequence of the free-block Lyapunov equation alone: an arbitrary outgoing vector need not lie in $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$. It says that the deterministic channel carrying target fluctuations into the free-block has finite covariance energy relative to the free-noise background.

To understand this condition intuitively, consider a finite-dimensional setting where the free-block state space is $\mathcal{H}_I = \mathbb{R}^{d-1}$ and the free stationary covariance $Q_0 \coloneqq \Sigma_{II}^{(0)}$ is strictly positive-definite. Since $Q_0 \succ 0$, its square root $Q_0^{1/2}$ is invertible, and its range is the entire space $\mathbb{R}^{d-1}$. In this case, the Cameron–Martin space is the full space $H_{Q_0} = \mathrm{Range}(Q_0^{1/2}) = \mathbb{R}^{d-1}$, and the Cameron–Martin energy norm $|u|_{Q_0} = \|Q_0^{-1/2}u\|_2$ is equivalent to the standard Euclidean norm. Consequently, the admissibility condition $a_{Ik} \in H_{Q_0}$ is trivially satisfied for any outgoing vector $a_{Ik} \in \mathbb{R}^{d-1}$.

However, if some direction in the free coordinates is noise-free under block decoupling (meaning Q_0 is positive semidefinite but singular), H_{Q_0} becomes a proper subspace of $\mathbb{R}^{d-1}$. In this case, the condition $a_{Ik} \in H_{Q_0}$ requires that the leakage vector a_{Ik} has no component in the null space of Q_0. Equivalently, it prevents the target coordinate from leaking fluctuations into a downstream coordinate that has zero background noise, which would otherwise result in infinite relative covariance energy.

In infinite dimensions, $\Sigma_{II}^{(0)}$ is trace-class and compact, so its eigenvalues decay to zero, making $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$ a dense but strictly proper subspace of $\mathcal{H}_I$. The condition $a_{Ik} \in H_{Q_0}$ requires the off-diagonal drift coupling coefficients to decay sufficiently fast relative to the background noise eigenvalues to keep the relative energy finite.

Under this assumption, invariance and exponential stability on H_{Q_0} imply:

$$
\begin{gathered}
(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik} \in H_{Q_0}, \\
|(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}|_{Q_0} \le \frac{M}{\lambda + \omega} |a_{Ik}|_{Q_0}.
\end{gathered}
\tag{17}
$$

Remark 8 (Constant Dependency in LMMSE Expansion). Consequently, the exact leakage cross-covariance $\Sigma_{\lambda,Ik}$ has finite Cameron–Martin energy norm and is $O(\lambda^{-3})$ in that norm. This is a condition on the deterministic response and covariance energy; it does not assert that infinite-dimensional Gaussian sample paths lie in H_{Q_0} almost surely.

3. Hilbert-Space Variational Schur Residual

3.1. **Variational Schur Subtraction and Energy Bounds.** In finite dimensions, the Schur complement of the bulk covariance block Σ_{II} is defined as $S_k \coloneqq \Sigma_{kk} - \Sigma_{kI}\Sigma_{II}^{-1}\Sigma_{Ik}$. Under stable and controllable dynamics, the bulk covariance Σ_{II} is positive-definite and invertible, so the subtraction is well-defined. However, in infinite dimensions, the stationary covariance operator is trace-class and compact, meaning its restriction $\Sigma_{\lambda,II}$ lacks a bounded global inverse. The appropriate operator-theoretic replacement is the shorted operator: for a non-negative block operator, the Schur complement is recovered as the quadratic form of the short to the target subspace, with the inverse term replaced by a variational supremum [16, 1]. To lift the Schur complement to infinite dimensions in the present singular-conditioning problem, we formulate the projection residual directly using covariance-energy forms.

Let $Q_0 \coloneqq \Sigma_{II}^{(0)}$ denote the fixed background bulk covariance under full decoupling ($\lambda = \infty$). We define the core Cameron–Martin space $H_{Q_0} \coloneqq \mathrm{Range}(Q_0^{1/2})$ and the core energy norm:

$$|u|_{Q_0} \coloneqq \|Q_0^{-1/2}u\| \quad \text{for all } u \in H_{Q_0}. \tag{18}$$

By Assumption 6, the off-diagonal drift coupling vector satisfies $a_{Ik} \in H_{Q_0}$, and the restricted drift generator $\mathcal{A}_{II}$ generates an exponentially stable semigroup on H_{Q_0}.

Under this assumption, the exact cross-covariance resolvent formula (derived in Section A) satisfies:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1}a_{Ik}. \tag{19}$$

This resolvent response yields the Cameron–Martin energy bound on the cross-covariance block:

$$|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}) \quad \text{as } \lambda \to \infty. \tag{20}$$

To define the infinite-dimensional projection residual, we introduce the variational Schur subtraction R_λ:

$$R_\lambda \coloneqq \sup_{\langle u, \Sigma_{\lambda,II}u\rangle > 0} \frac{|\langle u, \Sigma_{\lambda,Ik}\rangle|^2}{\langle u, \Sigma_{\lambda,II}u\rangle}. \tag{21}$$

Equivalently, if Σ_λ is viewed as a non-negative block operator on $\mathbb{C}\phi_k \oplus \mathcal{H}_I$, then (21) is the one-dimensional Anderson–Trapp shorted-operator subtraction, namely

$$\left\langle \phi_k, \mathcal{S}_{\mathbb{C}\phi_k}(\Sigma_\lambda)\phi_k \right\rangle = \Sigma_{\lambda,kk} - R_\lambda. \tag{22}$$

The restriction to vectors with $\langle u, \Sigma_{\lambda,II}u\rangle > 0$ simply removes null directions of the denominator; for a non-negative block operator, such directions do not change the supremum. Thus R_λ is not an ambient inverse computation but the energy removed by shorting to the target coordinate [1, 19].

Since the target noise channel and bulk noise channel are independent, the bulk state decomposes into independent background and target-driven components, meaning the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). Douglas range inclusion then guarantees that the variational subtraction is bounded:

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|_{Q_0}^2 = O(\lambda^{-6}). \tag{23}$$

3.2. **Hilbert-Space Schur Residual Scaling.** We now establish the main scaling theorem for the L^2-projection residual.

Theorem 9 (Hilbert Schur Residual Invariant)**.** *Under Assumptions 6 and 6, the L^2-projection residual satisfies:*

$$S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I} X_k\|_{L^2}^2 = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \tag{24}$$

where the remainder satisfies $0 \le R_\lambda = O(\lambda^{-6})$. In particular, the Schur LMMSE invariant is:

$$m_2^{\mathrm{Schur}} = \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda} = \frac{\varepsilon_0}{2}, \tag{25}$$

and the inverse projection residual scaling satisfies:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}). \tag{26}$$

This result depends only on covariance and orthogonal projection; it holds for any linear stationary model satisfying these assumptions, without Gaussianity.

Proof. See Section A for the full derivation. □

3.3. **Convergence of Finite-Dimensional Truncations.** Let $\Pi_n : \mathcal{H} \to \mathcal{H}_n := \operatorname{span}\{\phi_1, \dots, \phi_n\}$ be the n-dimensional Galerkin projection. Write $\mathcal{A}_n^{(\lambda)} := \Pi_n \mathcal{A}^{(\lambda)} \Pi_n$, $\mathcal{D}_n^{(\lambda)} := \Pi_n \mathcal{D}^{(\lambda)} \Pi_n$, and let $\Sigma_{n,\lambda}$ solve the truncated Lyapunov equation $\mathcal{A}_n^{(\lambda)} \Sigma_{n,\lambda} + \Sigma_{n,\lambda} (\mathcal{A}_n^{(\lambda)})^* + \mathcal{D}_n^{(\lambda)} = 0$.

Proposition 10 (Galerkin Consistency of the Schur Invariant)**.** *Under Assumption 6, let $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ be finite-rank orthogonal Galerkin projections preserving the target-bulk block decomposition:*

$$\Pi_n \phi_k = \phi_k, \qquad \Pi_n \mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \text{ strongly on } \mathcal{H}. \tag{27}$$

Assume the truncated regularized dynamics satisfy the uniform exponential stability and trace-class conditions stated in Proposition 18 of Section A. Then the truncated covariance operators converge in trace norm:

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0 \quad \text{as } n \to \infty, \tag{28}$$

and the truncated Schur invariant is exact at every level:

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2} \quad \text{for all } n \ge k. \tag{29}$$

Proof. See Section A for the dynamic and spectral support. □

4. Gaussian Conditional Extension and Finite-Part Functional

This section gives the Gaussian extension of the base theorem, using Radon Gaussian laws and disintegration to formulate conditioning under exact evidence and define the renormalized finite-part functional.

4.1. **Radon Gaussian Setup and Disintegration.**

Assumption 11 (Hilbert Gaussian Conditioning and Affine Mean Channel)**.** In addition to Assumptions 6 and 6, let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Gaussian Radon law on $\mathcal{H}$ with injective trace-class covariance. Let μ_{obs} be the canonical Gaussian disintegration member on the evidence hyperplane $X_k = x_k^*$. Suppose that a fixed deterministic forcing $b \in \mathcal{H}$ gives the stationary mean equation:

$$\mathcal{A}_{\text{cond}} m_\lambda + b - \lambda \Pi_k (m_\lambda - x_k^* \phi_k) = 0, \quad \Pi_k \mathcal{A}_{\text{cond}} = 0. \tag{30}$$

We define the target deterministic offset constant:

$$c_\lambda := \lambda(m_{\lambda,k} - x_k^*) = b_k, \tag{31}$$

where $m_{\lambda,k} := \langle m_\lambda, \phi_k \rangle$ and $b_k := \langle b, \phi_k \rangle$.

Under the Gaussian Radon law, the coordinate projection $\pi_k : \mathcal{H} \to \mathbb{R}$ is continuous, so the target variable X_k is a well-defined Borel random variable. The Polish structure of the Hilbert space guarantees the existence of a regular conditional probability distribution. By selecting the weakly continuous version of the conditional law at the single point $t = x_k^*$ (as detailed in Section B), we obtain the unique pointwise disintegration member μ_{obs}. Under this measure, the target coordinate satisfies the exact target zeros:

$$m_{\text{obs},k} - x_k^* = 0, \quad \text{Var}_{\text{obs}}(X_k) = 0 \quad \text{(exact)}. \tag{32}$$

4.2. **Global-Inverse-Free Measurable Predictor.** Let $C_\lambda := \Sigma_{\lambda,II}$ denote the Gaussian bulk covariance operator on $\mathcal{H}_I$. Because C_λ is trace-class and compact, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} := \text{Range}(C_\lambda^{1/2})$ and the corresponding energy norm.

The leakage cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm $\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} = O(\lambda^{-3})$. This range inclusion allows us to define the conditional predictor $\mu_{k|I}(X_I) := \mathbb{E}[X_k \mid X_I]$ as a measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{33}$$

which is the unique LMMSE predictor in L^2. The estimates in Section B imply that, for all sufficiently large λ, the bulk marginals $\mu_{\text{obs},I}$ and $\mu_{\lambda,I}$ are equivalent and have finite bulk relative entropy:

$$D_{\text{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \text{as } \lambda \to \infty. \tag{34}$$

4.3. **Gaussian Renormalized Finite-Part Functional.** Although the joint KL divergence diverges under the exact conditioning, we can define a renormalized finite-part action.

Definition 12 (Gaussian Renormalized Finite-Part Functional)**.** Whenever $S_{k,\lambda} > 0$ and $\mu_{\mathrm{obs},I}$ is equivalent to $\mu_{\lambda,I}$, we define the Gaussian renormalized finite-part functional of the conditioned law:

$$\begin{aligned}\mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda) :=& D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) \\ &+ \frac{1}{2} S_{k,\lambda}^{-1} \mathbb{E}_{\mu_{\mathrm{obs}}}\left[\left(X_k - \mu_{k|I}(X_I)\right)^2\right].\end{aligned} \tag{35}$$

This represents the finite part of the joint KL divergence under the conditional normalization scheme, where the Dirac and Gaussian normalization singularities are subtracted.

Weak scheme-independence. For an evidence level $t \in \mathbb{R}$, write μ_λ^t for the weakly continuous conditional law from Section B and set

$$\mathfrak{J}_\lambda(t) := \mathfrak{J}_\lambda(\mu_\lambda^t;\mu_\lambda).$$

Then, for any two evidence levels t_1, t_2 for which the finite part is defined,

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{(t_2 - m_{\lambda,k})^2 - (t_1 - m_{\lambda,k})^2}{2\Sigma_{\lambda,kk}}. \tag{36}$$

In the calibrated scaling $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$, if $b_i = \lambda(m_{\lambda,k} - t_i)$, this becomes

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{b_2^2 - b_1^2}{\varepsilon_0}.$$

Proof. Let $\Delta_t := t - m_{\lambda,k}$ and

$$\rho_\lambda := \frac{\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}}^2}{\Sigma_{\lambda,kk}} = 1 - \frac{S_{k,\lambda}}{\Sigma_{\lambda,kk}}.$$

The Gaussian entropy computation in Section B gives

$$D_{\mathrm{KL}}(\mu_{\lambda,I}^t \parallel \mu_{\lambda,I}) = \frac{1}{2}\frac{\Delta_t^2}{\Sigma_{\lambda,kk}}\rho_\lambda - \frac{1}{2}\log(1-\rho_\lambda) - \frac{1}{2}\rho_\lambda.$$

The conditional-residual term in (35) is

$$\frac{1}{2S_{k,\lambda}}\mathbb{E}_{\mu_\lambda^t}\left[(X_k - \mu_{k|I}(X_I))^2\right] = \frac{1}{2}\rho_\lambda + \frac{1}{2}\frac{\Delta_t^2}{\Sigma_{\lambda,kk}}(1-\rho_\lambda).$$

Adding these two identities cancels the $\rho_\lambda/2$ terms and yields

$$\mathfrak{J}_\lambda(t) = \frac{(t - m_{\lambda,k})^2}{2\Sigma_{\lambda,kk}} - \frac{1}{2}\log(1-\rho_\lambda).$$

The logarithmic term depends on the reference covariance and the conditioning scale, but not on the evidence value t. Hence it cancels in the difference between t_2 and t_1, proving (36). The same argument shows weak independence from the mollifying kernel: replacing the Gaussian Dirac mollifier in Theorem 23 by any centered approximate identity with finite second moment changes the extracted finite part only by an additive term depending on the kernel, λ, and $S_{k,\lambda}$, not on t. Therefore all evidence differences $\Delta\mathfrak{J}_\lambda$ are kernel-independent. $\square$

4.4. **Gaussian Main Theorem.** We now establish the main regularization and cancellation theorems for the finite-part functional.

Theorem 13 (Gaussian Finite-Part Regularization). *Under Assumption 11, there exists $\lambda_0 > 0$ such that, for $\lambda \geq \lambda_0$, the renormalized finite-part functional under the conditioned law μ_{obs} is well-defined and satisfies:*

$$\mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}). \tag{37}$$

In particular, the functional remains bounded at $O(1)$ and converges to:

$$\lim_{\lambda\to\infty} \mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0}. \tag{38}$$

Proof. See Section B for the full derivation. □

Corollary 14 (Exact Decoupled-Centered Cancellation). *Under Assumption 11, if the target coordinate is dynamically decoupled from the bulk ($a_{Ik} = 0$) and the evidence is centered at the stationary mean ($x_k^* = m_{\lambda,k}$), then:*

$$\mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = 0 \quad \textit{for every } \lambda > 0. \tag{39}$$

Proof. The decoupling implies $\Sigma_{\lambda,Ik} = 0$, so $D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = 0$. The centering makes the conditional residual zero under the conditioning, so both terms in (35) vanish. □

5. Elliptic Gaussian Field Example

We now record a standard elliptic Gaussian field example to show that the abstract Hilbert-space assumptions are nonempty and natural for covariance operators arising from stochastic evolution equations and elliptic Gaussian priors [8, 18, 23].

5.1. **Elliptic Gaussian field on $[0,1]$.** Let

$$L = -\frac{d^2}{dx^2} + \kappa^2$$

on $L^2(0,1)$ with Dirichlet boundary conditions and $\kappa > 0$. Then $\mathcal{A}_0 = -L$ with domain $H^2(0,1) \cap H_0^1(0,1)$ is dissipative, L is positive self-adjoint and sectorial, and e^{-tL} is an exponentially stable analytic contraction semigroup. The inverse covariance

$$Q_0 = L^{-1}$$

is compact, positive, self-adjoint, and trace class. With eigenfunctions $e_j(x) = \sqrt{2}\sin(\pi j x)$, its eigenvalues are

$$q_j = (\pi^2 j^2 + \kappa^2)^{-1}.$$

Thus Q_0 is a typical infinite-dimensional covariance operator: it is compact and trace class, has no bounded inverse on the ambient space, and carries its inverse only as a Cameron–Martin energy form [2, 21].

5.2. **Cameron–Martin range and covariance energy.** The Cameron–Martin space associated with Q_0 is

$$H_{Q_0} = \mathrm{Range}(Q_0^{1/2}) = H_0^1(0,1),$$

with equivalent energy norm

$$\|h\|_{Q_0}^2 = \langle h, Lh\rangle_{L^2} = \int_0^1 \left(|h'(x)|^2 + \kappa^2 |h(x)|^2\right) dx.$$

This identifies the variational Schur projection in a Sobolev-energy form, as in the standard covariance-energy representation of Gaussian Hilbert priors [2, 23]. The covariance inverse does not act as a bounded operator on $L^2(0,1)$; it acts only through the energy form on its Cameron–Martin domain.

5.3. **Spectral sensor conditioning.** Fix a target eigenmode e_k and define the observed coordinate

$$X_k = \langle X, e_k\rangle_{L^2}.$$

The unperturbed target variance is

$$\Sigma_{0,kk} = q_k = (\pi^2 k^2 + \kappa^2)^{-1}.$$

A regularized restoring drift on this coordinate produces the exact normalization

$$\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad R_\lambda = O(\lambda^{-6}),$$

under the block assumptions of Section 3. Off-diagonal bounded perturbations of the drift operator generate the cross-covariance vector entering the variational Schur residual. The resulting subtraction is evaluated by the Cameron–Martin energy form above, not by a global inverse on $L^2(0,1)$.

This example shows that the abstract theorem applies to a genuine Gaussian field. The residual is controlled by a Sobolev energy on $H_0^1(0,1)$ while the ambient covariance remains compact.

6. Discussion

We have shown that singular Gaussian coordinate conditioning in a separable Hilbert space admits an inverse-free Schur normal form. The main obstruction is the absence of a bounded inverse for compact covariance blocks. The variational Cameron–Martin projection is the target-coordinate shorted-operator subtraction evaluated in the covariance-energy scale, and it yields the residual estimate needed for the hard-conditioning limit. The scalar nature of the target coordinate is essential to the present proof. Three extensions mark the boundary between the theorem proved here and a genuinely finite-rank conditioning theory. We formalize these extensions as exact theorems below.

6.1. **Matrix algebraic extension.** For an m-dimensional observed subspace $E = \operatorname{span}\{\phi_1, \dots, \phi_m\}$, the scalar target block $\Sigma_{\lambda,kk}$ is replaced by the covariance matrix $\Sigma_{\lambda,EE} \in \mathbb{R}^{m\times m}$. The scalar calibration becomes a matrix Lyapunov balancing problem.

Theorem 15 (Matrix Lyapunov Calibration)**.** *Let E be an m-dimensional target subspace. Suppose the regularized drift and diffusion on E are given by $A_{\lambda,E} = -\lambda\Lambda_E$ and $D_{\lambda,E} = \frac{1}{\lambda}\Delta_E$, where $\Lambda_E, \Delta_E \in \mathbb{R}^{m\times m}$ are strictly positive definite matrices. Let $\Sigma_{\lambda,EE}$ be the unique positive definite solution to the matrix Lyapunov equation:*

$$A_{\lambda,E}\Sigma_{\lambda,EE} + \Sigma_{\lambda,EE}A^*_{\lambda,E} + D_{\lambda,E} = 0. \tag{40}$$

Then $\Sigma_{\lambda,EE}$ scales exactly as λ^{-2}. Specifically, the normalized matrix $\widetilde{\Sigma}_{EE} \coloneqq \lambda^2\Sigma_{\lambda,EE}$ is independent of λ and is the unique positive definite solution to the scale-free Lyapunov equation:

$$\Lambda_E\widetilde{\Sigma}_{EE} + \widetilde{\Sigma}_{EE}\Lambda^*_E = \Delta_E. \tag{41}$$

Proof. Substituting $A_{\lambda,E} = -\lambda\Lambda_E$ and $D_{\lambda,E} = \frac{1}{\lambda}\Delta_E$ into (40) yields:

$$-\lambda\Lambda_E\Sigma_{\lambda,EE} - \lambda\Sigma_{\lambda,EE}\Lambda^*_E + \frac{1}{\lambda}\Delta_E = 0.$$

Multiplying the entire equation by λ gives:

$$\Lambda_E(\lambda^2\Sigma_{\lambda,EE}) + (\lambda^2\Sigma_{\lambda,EE})\Lambda^*_E = \Delta_E.$$

Defining $\widetilde{\Sigma}_{EE} \coloneqq \lambda^2\Sigma_{\lambda,EE}$, we recover (41). Since Λ_E is positive definite, its spectrum lies in the open right half-plane, making $-\Lambda_E$ Hurwitz. Because Δ_E is positive definite, standard continuous Lyapunov theory guarantees that (41) admits a unique positive definite solution $\widetilde{\Sigma}_{EE}$. Therefore, the exact scaling $\Sigma_{\lambda,EE} = \lambda^{-2}\widetilde{\Sigma}_{EE}$ holds for all $\lambda > 0$, properly generalizing the scalar calibration condition $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$. □

6.2. **Operator-valued residuals.** For an m-dimensional target space E, the Schur residual must be a positive-semidefinite operator on E. Because the free-block covariance $\Sigma_{\lambda,II}$ is compact and lacks a bounded global inverse, the formal finite-dimensional matrix expression $R_{\lambda,E} = \Sigma_{\lambda,EI}\Sigma^{-1}_{\lambda,II}\Sigma_{\lambda,IE}$ must be rigorously replaced by the shorted-operator construction evaluated in the Cameron–Martin energy norm.

Theorem 16 (Operator-Valued Schur Residual)**.** *Let $\mathcal{H} = E \oplus \mathcal{H}_I$, with E finite-dimensional. Let Σ_λ be the non-negative trace-class covariance operator on $\mathcal{H}$ with blocks $\Sigma_{\lambda,EE}$, $\Sigma_{\lambda,EI}$, $\Sigma_{\lambda,IE}$, and $\Sigma_{\lambda,II}$. Define the shorted-operator residual $R_{\lambda,E}$ on E by its quadratic forms:*

$$\langle v, R_{\lambda,E}v\rangle_E = \sup_{\langle u,\Sigma_{\lambda,II}u\rangle>0} \frac{|\langle u, \Sigma_{\lambda,IE}v\rangle|^2}{\langle u, \Sigma_{\lambda,II}u\rangle}, \qquad v \in E. \tag{42}$$

Assume the structural range inclusion $\mathrm{Range}(\Sigma_{\lambda,IE}) \subset \mathrm{Range}(\Sigma_{\lambda,II}^{1/2}) =: H_{C_\lambda}$. *Then* $R_{\lambda,E}$ *is a well-defined bounded positive-semidefinite operator on* E*, uniquely characterized by:*

$$R_{\lambda,E} = (\Sigma_{\lambda,II}^{-1/2}\Sigma_{\lambda,IE})^*(\Sigma_{\lambda,II}^{-1/2}\Sigma_{\lambda,IE}). \tag{43}$$

Furthermore, if the cross-covariance satisfies the energy bound $\|\Sigma_{\lambda,IE}v\|_{H_{C_\lambda}} \leq M\lambda^{-3}\|v\|_E$ *for all* $v \in E$ *and some constant* $M > 0$*, then the operator norm of the residual satisfies* $\|R_{\lambda,E}\|_{\mathcal{L}(E)} = O(\lambda^{-6})$.

Proof. By the assumed range inclusion, for any $v \in E$, the vector $y_v := \Sigma_{\lambda,IE}v$ lies in $H_{C_\lambda} = \mathrm{Range}(\Sigma_{\lambda,II}^{1/2})$. Thus, there exists a unique element $w_v \in \overline{\mathrm{Range}(\Sigma_{\lambda,II}^{1/2})}$ such that $\Sigma_{\lambda,II}^{1/2}w_v = y_v$. We denote $w_v := \Sigma_{\lambda,II}^{-1/2}\Sigma_{\lambda,IE}v$.

For any $u \in \mathcal{H}_I$ with $\langle u, \Sigma_{\lambda,II}u\rangle > 0$, the ratio in (42) can be rewritten as:

$$\frac{|\langle u, \Sigma_{\lambda,II}^{1/2}w_v\rangle|^2}{\|\Sigma_{\lambda,II}^{1/2}u\|^2} = \frac{|\langle \Sigma_{\lambda,II}^{1/2}u, w_v\rangle|^2}{\|\Sigma_{\lambda,II}^{1/2}u\|^2}.$$

By the Cauchy–Schwarz inequality, this ratio is bounded above by $\|w_v\|^2$. The supremum is achieved by taking a sequence of vectors $\Sigma_{\lambda,II}^{1/2}u$ converging to w_v, yielding:

$$\langle v, R_{\lambda,E}v\rangle_E = \|w_v\|^2 = \|\Sigma_{\lambda,II}^{-1/2}\Sigma_{\lambda,IE}v\|^2 = \langle v, (\Sigma_{\lambda,II}^{-1/2}\Sigma_{\lambda,IE})^*(\Sigma_{\lambda,II}^{-1/2}\Sigma_{\lambda,IE})v\rangle_E.$$

Since this holds for all $v \in E$, it uniquely defines $R_{\lambda,E}$ as a positive-semidefinite matrix on E. Finally, the assumption $\|\Sigma_{\lambda,IE}v\|_{H_{C_\lambda}} \leq M\lambda^{-3}\|v\|_E$ means exactly that $\|w_v\| \leq M\lambda^{-3}\|v\|_E$. Therefore, $\langle v, R_{\lambda,E}v\rangle_E \leq M^2\lambda^{-6}\|v\|_E^2$, which implies the operator norm bound $\|R_{\lambda,E}\|_{\mathcal{L}(E)} \leq M^2\lambda^{-6} = O(\lambda^{-6})$. □

6.3. **General observation operators and misalignment.** The scalar proof in this paper uses a block decomposition aligned with a single observed coordinate. However, typical finite-rank observation operators $L : \mathcal{H} \to \mathbb{R}^m$ select a subspace E that does not commute with the free prior covariance, meaning the target-free split is not an eigenbasis split. This geometric misalignment generates cross-coupling that requires a precise resolvent analysis.

Theorem 17 (Misaligned Covariance Coupling)**.** *Let* P_E *be an orthogonal projection onto an* m*-dimensional observed subspace* E*, and let* $\mathcal{H}_I = E^\perp$*. Let the regularized dynamics be decomposed such that the target block matches Theorem 15 with* $A_{\lambda,E} = -\lambda\Lambda_E$*, the cross-noise is zero (*$D_{\lambda,IE} = 0$*), and the off-diagonal drift coupling is denoted by* $A_{\lambda,IE}$*. If* $A_{\lambda,IE}$ *maps* E *into the Cameron–Martin space* H_{Q_0} *of the free background covariance* Q_0 *on* $\mathcal{H}_I$*, and the restricted drift* A_{II} *generates an exponentially stable semigroup on* H_{Q_0}*, then the cross-covariance satisfies the exact identity:*

$$\Sigma_{\lambda,IE} = \int_0^\infty e^{tA_{II}}A_{\lambda,IE}\Sigma_{\lambda,EE}e^{tA_{\lambda,E}^*}\,dt. \tag{44}$$

Furthermore, the misaligned cross-covariance energy scales as $\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} = O(\lambda^{-3})$ *for any* $v \in E$.

Proof. The off-diagonal block of the global Lyapunov equation $\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0$ yields the Sylvester equation:

$$A_{II}\Sigma_{\lambda,IE} + \Sigma_{\lambda,IE}A^*_{\lambda,E} + A_{\lambda,IE}\Sigma_{\lambda,EE} = 0.$$

Since A_{II} generates an exponentially stable semigroup $e^{tA_{II}}$ and $A_{\lambda,E} = -\lambda\Lambda_E$ is Hurwitz, the unique solution is given by the integral representation (44).

To bound the Cameron–Martin norm, fix $v \in E$. We have $e^{tA^*_{\lambda,E}}v = e^{-\lambda t\Lambda^*_E}v$. By Theorem 15, $\Sigma_{\lambda,EE} = O(\lambda^{-2})$. Since $A_{\lambda,IE}$ maps into H_{Q_0} and E is finite-dimensional, there exists a constant C such that $\|A_{\lambda,IE}w\|_{H_{Q_0}} \leq C\|w\|_E$ for all $w \in E$. Using the uniform exponential stability of $e^{tA_{II}}$ on H_{Q_0}, namely $\|e^{tA_{II}}\|_{\mathcal{L}(H_{Q_0})} \leq Me^{-\omega t}$ for some $M \geq 1, \omega > 0$, we estimate:

$$\begin{aligned}
\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} &\leq \int_0^\infty \|e^{tA_{II}}\|_{\mathcal{L}(H_{Q_0})}\|A_{\lambda,IE}\Sigma_{\lambda,EE}e^{-\lambda t\Lambda^*_E}v\|_{H_{Q_0}}dt \\
&\leq \int_0^\infty Me^{-\omega t}C\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)}\|e^{-\lambda t\Lambda^*_E}\|_{\mathcal{L}(E)}\|v\|_E dt.
\end{aligned}$$

Let $\gamma > 0$ be the smallest eigenvalue of the positive definite matrix Λ^*_E, so $\|e^{-\lambda t\Lambda^*_E}\|_{\mathcal{L}(E)} \leq Ke^{-\lambda\gamma t}$. Then:

$$\begin{aligned}
\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} &\leq MCK\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)}\|v\|_E \int_0^\infty e^{-(\omega+\lambda\gamma)t}dt \\
&= MCK\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)}\|v\|_E \frac{1}{\omega+\lambda\gamma}.
\end{aligned}$$

Since $\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)} = O(\lambda^{-2})$ and $(\omega+\lambda\gamma)^{-1} = O(\lambda^{-1})$, we conclude that $\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} = O(\lambda^{-3})$. Combined with Theorem 16, this verifies that the shorted-operator residual maintains the $O(\lambda^{-6})$ scaling even in the presence of observation misalignment. □

The result is therefore deliberately limited to covariance and Gaussian-conditioning questions in the rank-one aligned setting. It does not assert a general non-Gaussian conditioning theory. For non-Gaussian stationary processes, the projection residual $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - R_\lambda$ still has a second-moment interpretation, but Radon disintegration, finite-part functionals, and measure-class behavior require separate hypotheses [3, 15]. Dynamical relaxation after conditioning, or transport distances between post-conditioning laws, belongs to a different problem. The present contribution is the operator-theoretic construction of the singular Gaussian conditioning residual itself.

7. Conclusion

We established a Hilbert-space framework for singular coordinate conditioning of Gaussian measures. The central object is an inverse-free variational Schur projection on the Cameron–Martin range, equivalently the one-dimensional target-coordinate form of the shorted-operator subtraction. It gives canonical normalization, an $O(\lambda^{-6})$ residual estimate, Galerkin stability, and a Radon Gaussian conditioning statement. The elliptic field example shows that the assumptions are natural for trace-class covariance operators arising from stochastic evolution equations and elliptic Gaussian priors [8, 18, 23].

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used Google DeepMind's Antigravity assistant in order to merge the LaTeX manuscripts, clean up cross-referencing tags, format mathematical notation, and edit the draft for style consistency. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Appendix A. Exact Block and Variational Schur Analysis

This appendix derives the linear invariants from the block Lyapunov equation and proves the strong convergence of Galerkin truncations. We do not use Laurent series or covariance inverses.

A.1. **Exact Target Covariance Block.** Write $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ with $\mathcal{H}_k = \mathbb{R}\phi_k$. The incoming target-coordinate block decoupling and the calibrated noise scale give the block operators:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & D_{II} \end{pmatrix}. \tag{45}$$

The block $\Sigma_{\lambda,Ik} \in \mathcal{H}_I$ denotes the vector representation of the target-to-bulk cross-covariance. The kk block of the Lyapunov equation is the scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0 \quad \Longrightarrow \quad \Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2}. \tag{46}$$

This equality holds exactly for all $\lambda > 0$.

A.2. **Exact Cross-Covariance Resolvent Formula.** The Ik block of the Lyapunov equation reduces to:

$$(\mathcal{A}_{II} - \lambda I)\Sigma_{\lambda,Ik} + a_{Ik}\Sigma_{\lambda,kk} = 0. \tag{47}$$

This yields the exact cross-covariance resolvent formula:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1}a_{Ik}. \tag{48}$$

A.3. **Cameron–Martin Energy Bound.** Let $Q_0 \coloneqq \Sigma_{II}^{(0)}$ be the background bulk covariance. The Cameron–Martin space $H_{Q_0} \coloneqq \mathrm{Range}(Q_0^{1/2})$ has energy norm $|u|_{Q_0} \coloneqq \|Q_0^{-1/2} u\|_{\mathcal{H}_I}$. Under Assumption 6, the restricted resolvent satisfies the energy bound:

$$|(\lambda I - \mathcal{A}_{II})^{-1} u|_{Q_0} \leq \frac{M}{\lambda + \omega} |u|_{Q_0} \tag{49}$$

for all $u \in H_{Q_0}$. It follows from (48) that the cross-covariance has finite energy norm and decays as:

$$|\Sigma_{\lambda,Ik}|_{Q_0} \leq \frac{\varepsilon_0 M}{2\lambda^2(\lambda + \omega)} |a_{Ik}|_{Q_0} = O(\lambda^{-3}). \tag{50}$$

A.4. **Variational Schur Residual Proof.** We define the variational Schur subtraction R_λ:

$$R_\lambda = \sup_{\langle u, \Sigma_{\lambda,II} u\rangle > 0} \frac{1}{\langle u, \Sigma_{\lambda,II} u\rangle} |\langle u, \Sigma_{\lambda,Ik}\rangle|^2. \tag{51}$$

Because the target and bulk noise channels are independent, the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). We then have:

$$0 \leq R_\lambda \leq |\Sigma_{\lambda,Ik}|_{Q_0}^2 \leq \frac{\varepsilon_0^2 M^2 |a_{Ik}|_{Q_0}^2}{4\lambda^4(\lambda + \omega)^2} = O(\lambda^{-6}). \tag{52}$$

The projection residual satisfies $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$, so substituting the exact target block and variational remainder bound yields:

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}), \qquad m_2^{\mathrm{Schur}} = \frac{\varepsilon_0}{2}, \tag{53}$$

which implies the inverse scaling:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0} \lambda^2 + O(\lambda^{-2}). \tag{54}$$

A.5. **Galerkin Truncation Convergence.** Here we prove Proposition 10. The proof has two layers: a dynamic Bochner-integral argument that gives trace-norm convergence of the truncated covariance operators, followed by a spectral Schur approximation that yields monotone convergence and an explicit rate.

Proposition 18 (Exact Dynamic Galerkin Convergence Support)**.** *Let* $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ *be finite-rank orthogonal Galerkin projections such that*

$$\Pi_n \phi_k = \phi_k, \qquad \Pi_n \mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \tag{55}$$

strongly on $\mathcal{H}$*. For each* $\lambda > 0$*, define the truncated regularized dynamics by*

$$\mathcal{A}_n^{(\lambda)} \coloneqq \Pi_n \mathcal{A}^{(\lambda)}|_{\mathcal{H}_n}, \qquad \mathcal{D}_n^{(\lambda)} \coloneqq \Pi_n \mathcal{D}^{(\lambda)} \Pi_n. \tag{56}$$

Assume that $e^{t\mathcal{A}_n^{(\lambda)}}x \to e^{t\mathcal{A}^{(\lambda)}}x$ *for every* $x \in \mathcal{H}$*, uniformly for* t *in compact intervals, and that the semigroups are uniformly exponentially stable:*

$$\|e^{t\mathcal{A}_n^{(\lambda)}}\| \le Me^{-\beta t}, \qquad \|e^{t\mathcal{A}^{(\lambda)}}\| \le Me^{-\beta t}, \tag{57}$$

with $M \ge 1$*,* $\beta > 0$ *independent of* n*. Assume also*

$$\mathcal{D}_n^{(\lambda)} \to \mathcal{D}^{(\lambda)} \quad \text{in } \mathcal{S}_1(\mathcal{H}). \tag{58}$$

Then the Galerkin stationary covariance operators

$$\Sigma_{n,\lambda} = \int_0^\infty e^{t\mathcal{A}_n^{(\lambda)}}\mathcal{D}_n^{(\lambda)}e^{t\mathcal{A}_n^{(\lambda)*}}\,dt \tag{59}$$

converge in trace norm:

$$\Sigma_{n,\lambda} \to \Sigma_\lambda \quad \text{in } \mathcal{S}_1(\mathcal{H}). \tag{60}$$

Moreover, because the Galerkin scheme preserves the target-bulk block decomposition, one has the exact finite-n *target calibration*

$$\Sigma_{n,\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2} \tag{61}$$

for every $n \ge k$*. If the off-diagonal drift coupling satisfies the uniform finite-energy condition*

$$a_{Ik}^{(n)} \in H_{Q_0^{(n)}}, \qquad \sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} < \infty, \tag{62}$$

where $Q_0^{(n)} := \Sigma_{n,II}^{(0)}$ *is the truncated background covariance, then the truncated Schur residual satisfies*

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - R_{n,\lambda}, \qquad 0 \le R_{n,\lambda} \le C\lambda^{-6}. \tag{63}$$

Consequently,

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2} \tag{64}$$

for every admissible Galerkin truncation $n \ge k$*.*

Proof. Write the difference of covariances as the Bochner integral

$$\Sigma_{n,\lambda} - \Sigma_\lambda = \int_0^\infty \left(e^{t\mathcal{A}_n^{(\lambda)}}\mathcal{D}_n^{(\lambda)}e^{t\mathcal{A}_n^{(\lambda)*}} - e^{t\mathcal{A}^{(\lambda)}}\mathcal{D}^{(\lambda)}e^{t\mathcal{A}^{(\lambda)*}}\right)dt. \tag{65}$$

Split the integrand in the trace norm:

$$\begin{aligned}\left\|e^{t\mathcal{A}_n}\mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\right\|_{\mathcal{S}_1} &\le \left\|e^{t\mathcal{A}_n}(\mathcal{D}_n - \mathcal{D})e^{t\mathcal{A}_n^*}\right\|_{\mathcal{S}_1} \\ &\quad + \left\|e^{t\mathcal{A}_n}\mathcal{D}e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\right\|_{\mathcal{S}_1}.\end{aligned} \tag{66}$$

(We omit the superscript (λ) for readability.)

The first term is bounded by $M^2e^{-2\beta t}\|\mathcal{D}_n - \mathcal{D}\|_{\mathcal{S}_1}$, which tends to zero by assumption. For the second term, we use the two-sided ideal convergence property of $\mathcal{S}_1(\mathcal{H})$ (Grümm's theorem; see [21]): if $L_n \to L$ strongly, $R_n^* \to R^*$ strongly, the two sequences are uniformly bounded, and $K \in \mathcal{S}_1(\mathcal{H})$, then $L_nKR_n \to LKR$ in $\mathcal{S}_1(\mathcal{H})$. Applying this with $L_n = e^{t\mathcal{A}_n}$, $R_n = e^{t\mathcal{A}_n^*}$,

$L = e^{t\mathcal{A}}$, $R = e^{t\mathcal{A}^*}$, and $K = \mathcal{D}$, the second term converges to zero for each t. Crucially, $R_n^* = e^{t\mathcal{A}_n} \to e^{t\mathcal{A}} = R^*$ is already covered by the same Trotter–Kato strong convergence assumption; no independent strong convergence of the adjoint semigroups is required. The exponential stability gives the integrable dominating bound

(67)
$$\left\| e^{t\mathcal{A}_n}\mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*} \right\|_{\mathcal{S}_1} \le M^2 e^{-2\beta t}\big(\|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1}\big) \le M^2 C_{\mathcal{D}}\, e^{-2\beta t},$$

where $C_{\mathcal{D}} := \sup_n \|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1} < \infty$ (the supremum is finite because $\mathcal{D}_n \to \mathcal{D}$ in trace norm). The dominated-convergence theorem for Bochner integrals then yields

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0. \tag{68}$$

Because Π_n preserves $\mathcal{H}_k \oplus \mathcal{H}_I$, the truncated blocks remain

$$\mathcal{A}_n^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik}^{(n)} & \mathcal{A}_{II}^{(n)} \end{pmatrix}, \qquad \mathcal{D}_n^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & \mathcal{D}_{II}^{(n)} \end{pmatrix}. \tag{69}$$

Thus the (kk)-block of the truncated Lyapunov equation is exactly

$$-2\lambda\, \Sigma_{n,\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0, \tag{70}$$

hence $\Sigma_{n,\lambda,kk} = \varepsilon_0/(2\lambda^2)$. The (Ik)-block gives

$$(\mathcal{A}_{II}^{(n)} - \lambda I)\Sigma_{n,\lambda,Ik} + a_{Ik}^{(n)}\Sigma_{n,\lambda,kk} = 0, \tag{71}$$

so

$$\Sigma_{n,\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II}^{(n)})^{-1} a_{Ik}^{(n)}. \tag{72}$$

Uniform Cameron–Martin stability (the finite-energy condition on $a_{Ik}^{(n)}$) gives $|\Sigma_{n,\lambda,Ik}|_{Q_0^{(n)}} = O(\lambda^{-3})$ uniformly in n. Therefore the variational Schur subtraction satisfies

$$0 \le R_{n,\lambda} \le |\Sigma_{n,\lambda,Ik}|^2_{Q_0^{(n)}} = O(\lambda^{-6}), \tag{73}$$

and hence

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}). \tag{74}$$

Multiplying by λ^2 and letting $\lambda \to \infty$ gives $m_2^{\mathrm{Schur},(n)} = \varepsilon_0/2$. □

Lemma 19 (Spectral Schur Compression Support)**.** *Fix $\lambda > 0$ and let $C_\lambda := \Sigma_{\lambda,II}$ denote the bulk covariance operator on $\mathcal{H}_I$. Let $\{(\sigma_j, e_j)\}_{j=1}^\infty$ be the eigenpairs of C_λ with $\sigma_1 \ge \sigma_2 \ge \cdots > 0$. Define the spectral compression projectors*

$$P_n^\lambda := \sum_{j=1}^n e_j \otimes e_j, \qquad n \ge 1. \tag{75}$$

Under Assumption 6 (Cameron–Martin stability), the leakage vector satisfies $\Sigma_{\lambda,Ik} \in \mathrm{Range}(C_\lambda^{1/2})$*, so the spectral Schur compression*

$$R_\lambda^{[n]} := \sum_{j=1}^{n} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \tag{76}$$

is finite for every n *and monotone increasing:*

$$R_\lambda^{[n]} \uparrow R_\lambda := \|\Sigma_{\lambda,Ik}\|_{CM}^2 \quad \text{as } n \to \infty. \tag{77}$$

Consequently the compressed Schur residuals satisfy

$$S_{k,\lambda}^{[n]} := \frac{\varepsilon_0}{2\lambda^2} - R_\lambda^{[n]} \quad \searrow \quad S_{k,\lambda} := \frac{\varepsilon_0}{2\lambda^2} - R_\lambda \quad \text{as } n \to \infty. \tag{78}$$

If, moreover, the weighted Cameron–Martin tail condition

$$\sum_{j\geq 1} j^{\,\eta-1} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq C\lambda^{-6}, \qquad \eta > 1, \tag{79}$$

holds, then the tail is explicitly bounded by

$$S_{k,\lambda}^{[n]} - S_{k,\lambda} = \sum_{j=n+1}^{\infty} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq C\, n^{1-\eta}\lambda^{-6}. \tag{80}$$

Proof. By Assumption 6, $\Sigma_{\lambda,Ik} \in H_{Q_0} \subseteq \mathrm{Range}(C_\lambda^{1/2})$. Hence $C_\lambda^{-1/2}\Sigma_{\lambda,Ik} \in \mathcal{H}_I$ and

$$R_\lambda^{[n]} = \sum_{j=1}^{n} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} = \|P_n^\lambda C_\lambda^{-1/2}\Sigma_{\lambda,Ik}\|_{\mathcal{H}_I}^2 \leq \|C_\lambda^{-1/2}\Sigma_{\lambda,Ik}\|_{\mathcal{H}_I}^2 = R_\lambda. \tag{81}$$

Since $P_n^\lambda \to I$ strongly and the terms are non-negative, $R_\lambda^{[n]} \uparrow R_\lambda$. The exact target block (46) gives $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$, so $S_{k,\lambda}^{[n]} = \varepsilon_0/(2\lambda^2) - R_\lambda^{[n]} \searrow S_{k,\lambda}$. Using (50), $R_\lambda = O(\lambda^{-6})$, whence $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - O(\lambda^{-6})$ and $m_2^{\mathrm{Schur}} = \varepsilon_0/2$.

Under the weighted tail condition, the remainder after n terms satisfies

$$\sum_{j=n+1}^{\infty} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq n^{1-\eta} \sum_{j=n+1}^{\infty} j^{\,\eta-1} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq C\, n^{1-\eta}\lambda^{-6}, \tag{82}$$

which gives the asserted rate. □

Appendix B. Gaussian Conditional Finite-Part Analysis

This appendix derives the Radon Gaussian conditioning, measurable conditional predictor, bulk relative entropy bounds, and the finite-part regularization theorem via Dirac mollification.

B.1. **Radon Gaussian Setup and Pointwise Disintegration.** Let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Radon Gaussian measure on the separable Hilbert space $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$. The trace-class property of Σ_λ ensures its existence as a Radon measure [2]. Let $\pi_k : \mathcal{H} \to \mathbb{R}$ be the projection $\pi_k(X) = \langle X, \phi_k\rangle = X_k$.

Lemma 20 (Continuous Disintegration and Exact Target Zeros)**.** *For each $t \in \mathbb{R}$, there exists a unique weakly continuous family of Gaussian Radon measures $\mu_\lambda^t = \mathcal{N}(m_\lambda^t, \Sigma_\lambda^t)$ supported on the hyperplane $X_k = t$, where:*

$$m_\lambda^t := \begin{pmatrix} t \\ m_{\lambda,I} + \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(t - m_{\lambda,k}) \end{pmatrix}, \tag{83}$$

$$\Sigma_\lambda^t := \begin{pmatrix} 0 & 0 \\ 0 & \Sigma_{\lambda,II} - \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI} \end{pmatrix}. \tag{84}$$

Under the conditioned law $\mu_{\mathrm{obs}} := \mu_\lambda^{x_k^*}$, the target coordinate satisfies the exact target zeros:

$$\mathbb{E}_{\mathrm{obs}}[X_k] - x_k^* = 0, \quad \mathrm{Var}_{\mathrm{obs}}(X_k) = 0. \tag{85}$$

Proof. Since $\mathcal{H}$ is Polish, regular conditional probabilities exist. The stated Gaussian family is the continuous disintegration along the scalar observation map; its pointwise continuous version is unique [17]. The continuity of $t \mapsto m_\lambda^t$ in $\mathcal{H}$ and constancy of Σ_λ^t yield weak continuity of this family (see also [2]). Evaluating at $t = x_k^*$ gives $\mu_{\mathrm{obs}} = \mu_\lambda^{x_k^*}$, which has target component fixed at x_k^* almost surely. □

B.2. **Measurable Conditional Predictor.** Let $C_\lambda := \Sigma_{\lambda,II}$ be the bulk covariance on $\mathcal{H}_I$. Since C_λ is trace-class, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} := \mathrm{Range}(C_\lambda^{1/2})$.

Lemma 21 (Measurable Conditional Predictor)**.** *The cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm:*

$$\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} \le |\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}). \tag{86}$$

Furthermore, the conditional mean predictor $\mu_{k|I}(X_I) := \mathbb{E}_{\mu_\lambda}[X_k \mid X_I]$ is represented by the measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{87}$$

where $W_{\Sigma_{\lambda,Ik}}$ is the $L^2(\mu_{\lambda,I})$-limit of the continuous linear functionals on $\mathcal{H}_I$. For all sufficiently large λ, the conditional variance is strictly positive:

$$\mathrm{Var}(X_k \mid X_I) = \Sigma_{\lambda,kk} - \|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}}^2 = S_{k,\lambda} > 0. \tag{88}$$

Proof. Since $\Sigma_{\lambda,II} \succeq Q_0$, Douglas range inclusion [9] yields $\mathrm{Range}(Q_0^{1/2}) \subseteq \mathrm{Range}(\Sigma_{\lambda,II}^{1/2}) = H_{C_\lambda}$ with the contraction inequality

$$\|\Sigma_{\lambda,II}^{-1/2} v\|_{\mathcal{H}_I} \le \|Q_0^{-1/2} v\|_{\mathcal{H}_I} \quad \text{for all } v \in \mathrm{Range}(Q_0^{1/2}).$$

Hence, $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ and the norm bound follows directly from (50). The range inclusion $\Sigma_{\lambda,Ik} \in \mathrm{Range}(\Sigma_{\lambda,II}^{1/2})$ guarantees that the Wiener functional $W_{\Sigma_{\lambda,Ik}}$ is well-defined as a measurable function on $(\mathcal{H}_I, \mu_{\lambda,I})$ [2]. The conditional variance is the projection residual. By Theorem 9,

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}) > 0$$

for all sufficiently large λ. □

B.3. **Bulk Relative-Entropy Estimate.**

Lemma 22 (Feldman–Hájek Equivalence and Entropy Decay)**.** *There exists $\lambda_0 > 0$ such that, for every $\lambda \geq \lambda_0$, the bulk marginals $\mu_{\mathrm{obs},I}$ and $\mu_{\lambda,I}$ are equivalent Gaussian Radon measures on $\mathcal{H}_I$, with relative entropy:*

$$\begin{aligned} 2D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = \frac{(x_k^* - m_{\lambda,k})^2}{\Sigma_{\lambda,kk}}\rho_\lambda \\ - \log(1-\rho_\lambda) - \rho_\lambda, \end{aligned} \tag{89}$$

where the eigenvalue $\rho_\lambda := \|\Sigma_{\lambda,Ik}\|^2_{H_{C_\lambda}}/\Sigma_{\lambda,kk} = 1 - S_{k,\lambda}/\Sigma_{\lambda,kk} < 1$. In particular, we have $\rho_\lambda = O(\lambda^{-4})$ and:

$$D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \text{as } \lambda \to \infty. \tag{90}$$

Proof. Under disintegration, $\mu_{\mathrm{obs},I}$ is a Gaussian Radon law on $\mathcal{H}_I$ with mean $m_{\mathrm{obs},I} = m_{\lambda,I} + \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(x_k^* - m_{\lambda,k})$ and covariance $\Sigma_{\mathrm{obs},II} = C_\lambda - \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI}$. The covariance operators differ by the rank-one operator $T_\lambda = \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI}$. Since $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$, the relative covariance operator is a rank-one perturbation of the identity with its only non-unit eigenvalue equal to $1-\rho_\lambda$. By Lemma 21 and the exact target covariance, $\rho_\lambda = O(\lambda^{-4})$. Choose λ_0 so that $\rho_\lambda < 1$ for $\lambda \geq \lambda_0$. The Feldman–Hájek theorem [10, 2] then guarantees equivalence of the measures, and the relative entropy is:

$$\begin{aligned} 2D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = \|C_\lambda^{-1/2}(m_{\mathrm{obs},I} - m_{\lambda,I})\|^2_{\mathcal{H}_I} \\ - \log(1-\rho_\lambda) - \rho_\lambda. \end{aligned}$$

The mean-shift identity in the lemma follows from

$$m_{\mathrm{obs},I} - m_{\lambda,I} = \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(x_k^* - m_{\lambda,k}).$$

Assumption 11 gives $m_{\lambda,k} - x_k^* = O(\lambda^{-1})$, so that term is $O(\lambda^{-4})$. Finally,

$$-\log(1-\rho_\lambda) - \rho_\lambda = O(\rho_\lambda^2) = O(\lambda^{-8}),$$

which proves the asserted entropy decay. □

B.4. **Mollified Divergence and Finite-Part Regularization Proof.** To regularize the singular joint KL divergence, define a law with bulk marginal $\mu_{\mathrm{obs},I}$ and an independent mollified target conditional:

$$\nu_{\lambda,\delta} \coloneqq \mu_{\mathrm{obs},I} \otimes \mathcal{N}(x_k^*, \delta^2) \tag{91}$$

for $\delta > 0$, identifying $\mathcal{H}$ with $\mathcal{H}_I \times \mathcal{H}_k$ in this display.

Proposition 23 (Mollification Limit)**.** *For $\lambda \geq \lambda_0$ as in Lemma 22, define*

$$C_{\lambda,\delta} \coloneqq \frac{1}{2}\left(\log \frac{S_{k,\lambda}}{\delta^2} - 1\right), \tag{92}$$

Then

$$\lim_{\delta \downarrow 0} \left[D_{\mathrm{KL}}(\nu_{\lambda,\delta} \parallel \mu_\lambda) - C_{\lambda,\delta}\right] = \mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda). \tag{93}$$

Proof. Put $r_\lambda(X) \coloneqq X_k - \mu_{k|I}(X_I)$. The entropy chain rule and the scalar Gaussian formula give

$$\begin{aligned} D_{\mathrm{KL}}(\nu_{\lambda,\delta} \parallel \mu_\lambda) &= D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) + C_{\lambda,\delta} \\ &\quad + \frac{\delta^2}{2S_{k,\lambda}} + \frac{1}{2S_{k,\lambda}} \mathbb{E}_{\mu_{\mathrm{obs}}}[r_\lambda(X)^2]. \end{aligned} \tag{94}$$

Here $X_k = x_k^*$ under μ_{obs}. Since $S_{k,\lambda} > 0$ for $\lambda \geq \lambda_0$, the term $\delta^2/(2S_{k,\lambda})$ vanishes as $\delta \downarrow 0$. The remaining terms equal Definition 12. □

Proof of Theorem 13. Let $\Delta_\lambda \coloneqq x_k^* - m_{\lambda,k}$ and $Z_\lambda \coloneqq W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I})$. The conditional Gaussian formulas give

$$\mathbb{E}_{\mathrm{obs}}[Z_\lambda] = \rho_\lambda \Delta_\lambda, \tag{95}$$

$$\mathrm{Var}_{\mathrm{obs}}(Z_\lambda) = \Sigma_{\lambda,kk}\rho_\lambda(1 - \rho_\lambda). \tag{96}$$

Using $X_k = x_k^*$ under μ_{obs} and $S_{k,\lambda} = \Sigma_{\lambda,kk}(1 - \rho_\lambda)$, we obtain

$$\begin{aligned} &\frac{1}{2S_{k,\lambda}} \mathbb{E}_{\mu_{\mathrm{obs}}}[(X_k - \mu_{k|I}(X_I))^2] \\ &\quad = \frac{\rho_\lambda}{2} + \frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}}(1 - \rho_\lambda). \end{aligned} \tag{97}$$

Here

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2}{\varepsilon_0}, \qquad \rho_\lambda = O(\lambda^{-4}).$$

The conditional term is therefore $b_k^2/\varepsilon_0 + O(\lambda^{-4})$. Adding Lemma 22 proves

$$\mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}),$$

and hence the stated limit. □

B.5. **Exact Decoupled-Centered Consequence.** If the target coordinate is dynamically decoupled ($a_{Ik} = 0$), then $\Sigma_{\lambda,Ik} = 0$, $S_{k,\lambda} = \Sigma_{\lambda,kk} > 0$, and the bulk marginals coincide for every $\lambda > 0$. If the evidence is also centered ($x_k^* = m_{\lambda,k}$), both terms in Definition 12 vanish. Hence $\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = 0$ exactly for every $\lambda > 0$.

Appendix C. Numerical Diagnostics and Galerkin Verification

This appendix presents arbitrary-precision numerical validations of the variance normalization, Schur residual scaling, finite-part asymptotics, and Galerkin convergence rates, computed in 256-bit `BigFloat` to isolate the structural invariants from floating-point collapse.

C.1. **Truncation Convergence.** We illustrate the Galerkin convergence of Proposition 10 against a finite reference on an admissible model. The Hilbert space is $\ell^2(\mathbb{N})$ with coordinate basis $\{\phi_j\}$. The drift is $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where $\mathcal{A}_0\phi_j = -(1 + 0.01j^2)\phi_j$ and $\mathcal{K}\phi_1 = \sum_{j\geq 2} 12\, j^{-3}\phi_j$ is a bounded rank-one nilpotent perturbation. The noise is $\mathcal{D}\phi_j = j^{-3/2}\phi_j$, which is trace-class. The regularized conditioning acts at $k = 1$ with $\varepsilon_0 = 1$. Truncated covariances are compared against a finite reference $N_{\text{ref}} = 1500$.

Table 1. Truncation convergence diagnostics for Experiment C.1.

Diagnostic	Value	Expected
$\Vert\mathcal{D} - \Pi_n\mathcal{D}\Pi_n\Vert_{\mathcal{S}_1}$ slope	-0.499	$\to 0$
$\Vert\Sigma_n - \Sigma_{\text{ref}}\Vert_{\mathcal{S}_1}$ slope	-2.683	$\to 0$
$\Vert\Sigma_{n,\lambda} - \Sigma_{\text{ref},\lambda}\Vert_{\mathcal{S}_1}$ slope	-2.683	$\to 0$
$\vert S_{1,n,\lambda} - S_{1,\text{ref},\lambda}\vert$ slope	-5.504	$\to 0$
$m_2^{\text{Schur},(1000)}$ (extrapolated)	0.5000000	0.5
$\vert m_2^{\text{Schur},(1000)} - \varepsilon_0/2\vert$	6.87×10^{-14}	0

C.2. **Canonical Scaling Diagnostic.** We validate the canonical normalization of Proposition 2 in its decoupled reference channel. With incoming target-coordinate blocks and off-diagonal drift coupling set to zero, the target covariance is exactly $\Sigma_{\lambda,22} = d_\lambda/(2\lambda)$. Four candidate target-diffusion families are tested:

1. **Canonical:** $d_\lambda = \varepsilon_0/\lambda$.
2. **Slow:** $d_\lambda = \varepsilon_0$.
3. **Fast:** $d_\lambda = \varepsilon_0/\lambda^2$.
4. **Perturbed:** $d_\lambda = (\varepsilon_0/\lambda)(1 + \lambda^{-2})$.

All evaluations are performed in Julia using 256-bit `BigFloat` arithmetic. A probe at $\lambda = 2^{100}$ resolves the nonzero finite-scale defect of the perturbed family.

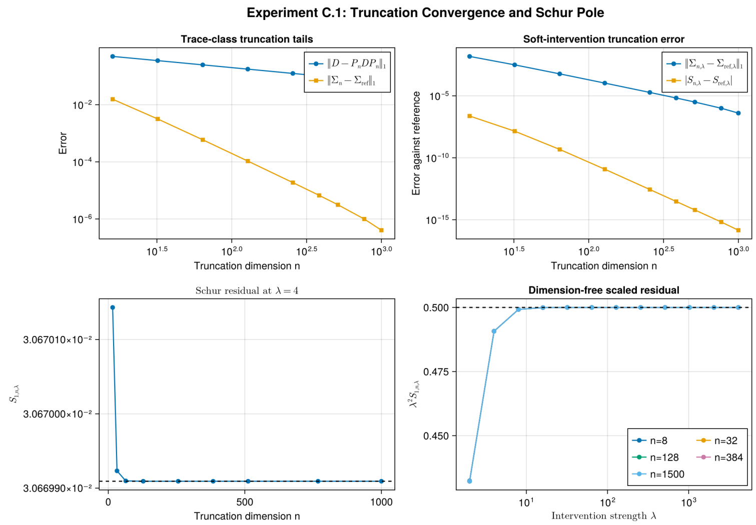

FIGURE 1. Experiment C.1. Trace-norm and residual errors decay in n; the extrapolated $m_2^{\mathrm{Schur},(1000)}$ differs from $\varepsilon_0/2$ by 6.87×10^{-14}.

TABLE 2. Exact-calibration ablation. Only the canonical family satisfies the exact finite-λ covariance condition.

Family	$\lambda d_\lambda/\varepsilon_0$	defect $\lvert 2\lambda^2\Sigma_{\lambda,kk}/\varepsilon_0 - 1\rvert$
Canonical	1	0
Slow	λ	$\lambda - 1 \to \infty$
Fast	λ^{-1}	$\lvert 1 - \lambda^{-1}\rvert \to 1$
Perturbed	$1 + \lambda^{-2}$	λ^{-2}

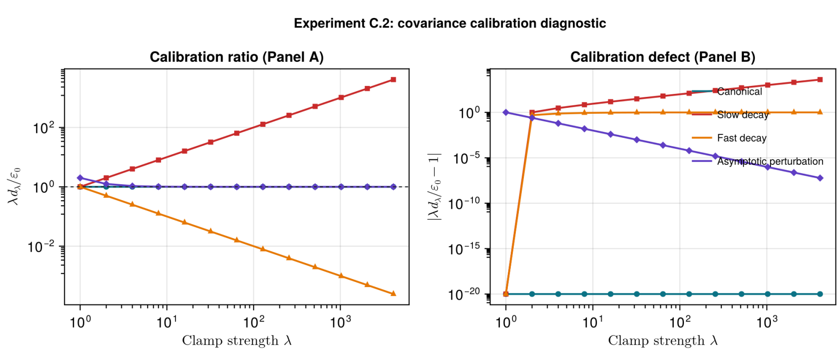

FIGURE 2. Experiment C.2. The perturbed family has the correct limiting ratio but fails the exact finite-λ covariance condition. Exact zero defects are displayed at 10^{-20}.

C.3. **Gaussian Finite-Part Functional in a Three-Node Chain.** We evaluate the Gaussian renormalized finite-part functional $\mathfrak{J}_\lambda$ on a three-node linear Ornstein–Uhlenbeck SDE:

$$X_1 \to X_2 \to X_3, \tag{98}$$

where X_2 is the target coordinate subject to regularized conditioning of strength $\lambda \in \{2^1, \dots, 2^{12}\}$, with $x_2^* = 0$ and $\varepsilon_0 = 1$. The drift and diffusion operators are:

$$A_\lambda = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & a_{32} & -1 \end{pmatrix}, \qquad D_\lambda = \operatorname{diag}(1, \varepsilon_0/\lambda, 1). \tag{99}$$

The target mean forcing b_2 controls the target mean shift $m_{\lambda,2} = b_2/\lambda$. We evaluate three Gaussian cases:

(1) **Case A (Decoupled-centered):** $a_{32} = 0, b_2 = 0$. Here $\mathfrak{J}_\lambda = 0$ exactly for all λ.
(2) **Case B (Coupled-centered):** $a_{32} = 1, b_2 = 0$. The coupling causes leakage $\Sigma_{23} \neq 0$, but the centering gives $\mathfrak{J}_\lambda = O(\lambda^{-4}) \to 0$.
(3) **Case C (Coupled-off-centered):** $a_{32} = 1, b_2 = 1$. The deterministic forcing causes mean shift $m_{\lambda,2} = 1/\lambda$, yielding $\mathfrak{J}_\lambda \to b_2^2/\varepsilon_0 = 1$.

All evaluations are performed in Julia using 256-bit `BigFloat` arithmetic, showing that the Schur residual $\lambda^2 S_{2,\lambda}$ converges to $1/2$ in all cases, while $\mathfrak{J}_\lambda$ exhibits the exact zero, $O(\lambda^{-4})$ decay, and convergence to 1 behavior respectively.

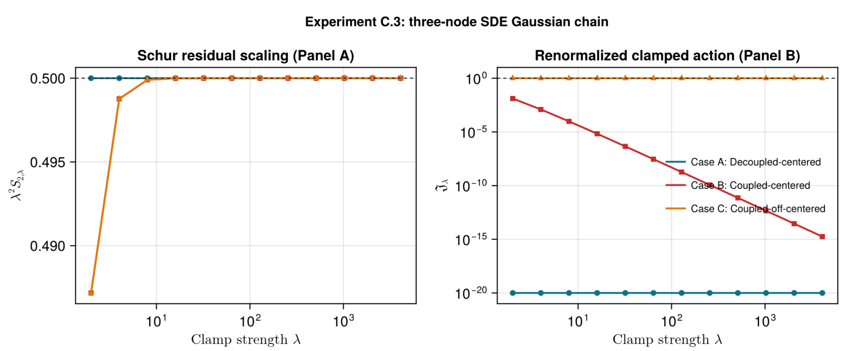

FIGURE 3. Experiment C.3. The panels show $\lambda^2 S_{2,\lambda} \to 0.5$ and the three finite-part regimes: exact zero, $O(\lambda^{-4})$ decay, and convergence to 1. Exact zeros are displayed at 10^{-20}.

C.4. **Numerical Precision Audit.** At $n = 256$, we recompute representative C.1 Schur residuals using IEEE 754 `Float64` and 256-bit Julia `BigFloat` arithmetic with the same covariance equation and Schur solve.

TABLE 3. Precision audit at $n = 256$; S_{256} uses 256-bit `BigFloat`.

λ	$\kappa_2(\Sigma_{II})$	$\lvert\lambda^2 S - \frac{1}{2}\rvert$	$\lvert S_{64} - S_{256}\rvert/\lvert S_{256}\rvert$
4	2.97×10^6	3.45×10^{-1}	6.50×10^{-16}
64	9.14×10^5	1.13×10^{-3}	3.48×10^{-17}
4096	9.14×10^5	9.20×10^{-10}	5.54×10^{-17}

For the five-point extrapolation of $m_2^{\text{Schur},(256)}$, `Float64` and `BigFloat` differ by 9.98×10^{-17}; finite-λ remainder, not rounding, governs the observed distance to the limiting constant.

Remark 24 (Block-decoupling error suppression). The sub-machine-epsilon agreement between `Float64` and `BigFloat` in Table 3 is not coincidental; it is a direct consequence of the block-triangular structure established in this paper. Under the block-decoupled conditioning, the drift matrix takes the form

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & A_{II} \end{pmatrix}, \tag{100}$$

so the operator Lyapunov equation decomposes into three sub-problems:

(1) The target block $\Sigma_{\lambda,kk} = d_\lambda/(2\lambda)$ is solved by a single scalar division, incurring at most $\frac{1}{2}\epsilon_{\text{machine}}$ relative rounding error (IEEE 754 round-to-nearest).
(2) The cross-covariance $\Sigma_{\lambda,Ik}$ satisfies $|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3})$ by (20), so the absolute rounding error in the cross block is $O(\lambda^{-3}\epsilon_{\text{machine}})$.
(3) The free block $\Sigma_{\lambda,II}$ is a standard Lyapunov solution with absolute error $O(\epsilon_{\text{machine}}\|\Sigma_{\lambda,II}\|)$.

When the Schur complement is assembled as $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$, the subtracted variational term satisfies $R_\lambda = O(\lambda^{-6})$ (Theorem 9), which is negligible compared to $\Sigma_{\lambda,kk} = O(\lambda^{-2})$. The rounding error in $S_{k,\lambda}$ is therefore dominated by the error in $\Sigma_{\lambda,kk}$ alone, yielding a Schur-level rounding budget of $O(\epsilon_{\text{machine}} \cdot \Sigma_{\lambda,kk})$ rather than the naïve $O(\epsilon_{\text{machine}} \cdot \|\Sigma_{\lambda,II}^{-1}\| \cdot \|\Sigma_{\lambda,Ik}\|^2)$ that one would expect without block decoupling.

In short, the block-triangular structure converts what would be a catastrophically ill-conditioned Schur computation into a well-conditioned scalar computation plus negligible corrections. The observed `Float64`–`BigFloat` agreement at $O(10^{-17})$ is the numerical signature of this structural decoupling.